

CHAPTER 1

Introduction and Overview

Books Recommended

Name: Data Structures (Schaum's Outline Series)

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Basic Terminology

- **Data Structure:**

- The logical or mathematical model of a particular organization of data.
- Data structure is a particular way of storing and organizing data in a computer so that it can be used efficiently.

- **Why Data Structure?**

- They allow you to store data so that they can be used in a program.
- Also allows for more efficient use of memory and runtime.
- Linear and Nonlinear data structures are two types.

NEED OF DATA STRUCTURE

- For example, list of names of months in a year, We can see that names of months have a linear relationship between them and thus can be stored in a sequential location, using an Array.
- Similarly, think of a set of data that represents location of historical places in a country. For each historical place the location is given by country name followed by state name followed by city name, and then followed by the historical place name.
- We can see that such data form a hierarchical relationship between them and must be represented in the memory using a hierarchical type of data structure, such as Tree.
- The above two examples clearly identify the usefulness of a data structure.

Basic Terminology

Linear Data Structure

- In linear data structure, member elements form a sequence. Such linear structures can be represented in memory by using one of the two basic strategies
 - By having relationship between the elements represented by means of sequential memory locations. These linear structures are called **arrays, stack, queue**.
 - By having relationship between the elements represented by pointers. These structures are called **linked lists**.

Basic Terminology

Non-Linear Data Structure

- In non-linear data structure, member elements may be connected with two or more other elements in a non-linear arrangement. Example: Tree, Graph.

Array

- The simplest type of data structure is linear array or 1D array.
- It is a list of a finite number n of similar data elements referenced respectively by a set of n consecutive numbers usually $1, 2, 3, \dots, n$.
- The element looks like $A_1, A_2, A_3, \dots, A_n$ or
 $A(1), A(2), A(3), \dots, A(n)$ or
 $A[1], A[2], A[3], \dots, A[n]$.

Linked List

- A linked list is linear collection of data elements where the linear order is maintained by means of pointers.
- Each node is divided into two parts: information part and link field.
- Information part contains information.
- Link field contains the address of the next node in the list.

Stack

- A stack is a list of elements in which elements may be inserted or deleted only at one end, called the Top of the stack.

Ex: Stack of dishes.

- It is a LIFO (Last in First out) system.
- PUSH is the term used to insert an element into a stack.
- POP is the term used to delete an element from a stack.

Queue

- A queue is a linear list of elements in which an elements can be deleted only at one end, called the 'front' and inserted only at the other end, called the rear.

Ex: People waiting in the train ticket counter.

- It is FIFO (First in First out) system.
- Another type of queue is Dequeue.
- Another is Priority queue.

Tree

- A binary tree T is defined as a finite set of elements, called the nodes, such that
 - (i) T is empty (called a null tree or empty tree).
or
 - (ii) T contains a distinguished node R , called the root of T , and the remaining nodes of T form an ordered pair of disjoint binary trees T_1 and T_2
- It is a nonlinear data structure.
- Some other types of Tree are:
 - (i) Complete Binary Tree
 - (ii) Extended Binary Tree (2 Tree) and
 - (iii) Binary Search Tree

Graph

- A Graph G (Undirected) consists of two things: $G=(V, E)$
 - i. A set V of elements called nodes (or points or vertices).
 - ii. A set E of edges such that each edge e in E is identified with a unique (unordered) pairs $[u, v]$ of nodes in V , denoted by $e=[u, v]$.
- It is also a nonlinear data structure.
- If $e=(v, w)$ is an edge with vertices u and w , then v and w are said to lie on e , and e is said to be incident with v and w .
- Another type of Graph is Directed Graph.